ITA0329-Mobile Computing

Co2-Assessment 2

Figma Designs

Date: 24-10-2026

Name: Duddela Harika

Reg No: 192324301

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

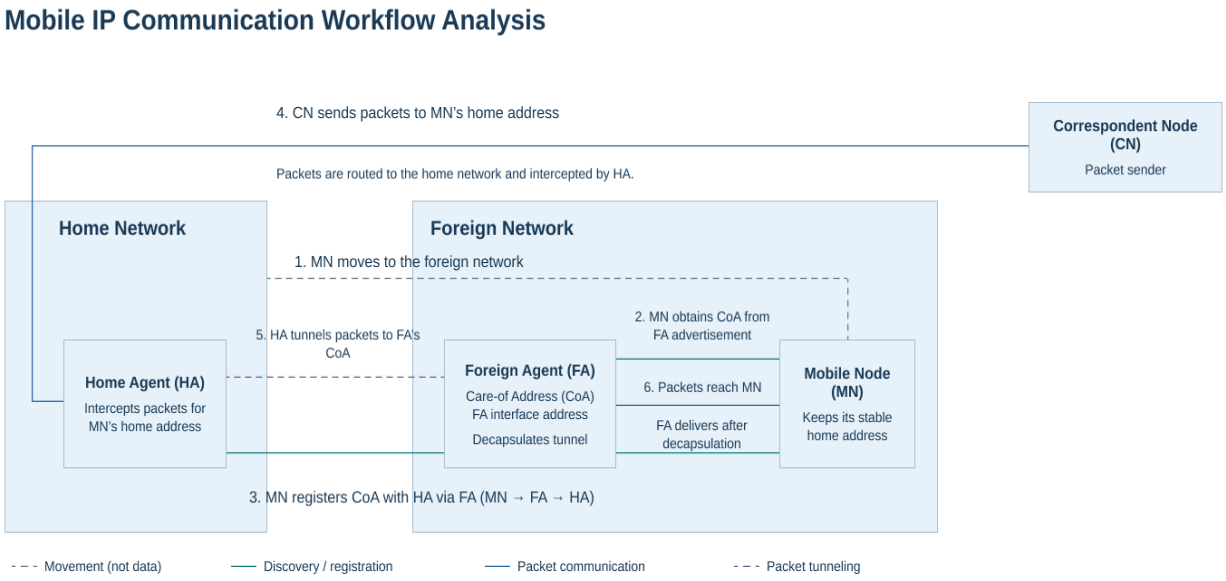
Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design.
- GitHub repository containing exported images/PDF and documentation (README).

Figma:



Communication Analysis

Component purposes

HA: Maintains the home-address → CoA binding and tunnels packets.

FA: Advertises its CoA, relays registration, and delivers packets to MN.

MN: Moves between networks while retaining its home address.

CoA: FA's advertised interface address; the tunnel endpoint, not a router.

CN: Sends packets addressed to MN's home address.

Packet delay and loss: Triangular routing (CN → HA → FA → MN) and tunnel overhead add delay. Handoff may interrupt delivery and lose packets.

Mobility effects: MN keeps a stable home address; HA updates its CoA binding after successful registration as MN changes networks.

Latency improvements: Use faster agent discovery and registration, buffering during handoff, and route optimization where supported.

2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.

Figma:

